V-OS

A graphical operating system for the Amstrad CPC 6128

Manual

Made from docs/manual/en/*.md and sdk/README.en.md by tools/mkdocs.py (make docs).

Contents

V-OS is a graphical operating system for the Amstrad CPC 6128, written from scratch in Z80 assembly. It has:

- a menu bar, draggable windows, dialogs and buttons, used with the keyboard or an AMX mouse
- three views for files and programs (Desktop, Launcher, Commander) and the settings (Setup)
- six languages as plugins: English, Greek, French, Spanish, German and Italian
- programs that run side by side and resume where they left off, a desk accessory and programs in the background
- Studio, a Scratch-style block editor that runs on the CPC itself
- an API with 42 calls and an SDK for programs in assembly

This manual is also in Greek (../00-index.md), French (../fr/00-index.md), Spanish (../es/00-index.md), German (../de/00-index.md) and Italian (../it/00-index.md).

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1. Getting started

What you need

- An Amstrad CPC 6128 or an emulator of it. V-OS is tested on Caprice32.
- The V-OS disc in drive A:, that is, the `build/vos.dsk` that `make` builds.
- Optionally:
  - an AMX mouse
  - a second drive (B:)
  - a memory expansion (e.g. 512K), for more programs at once

Starting up

From BASIC:

```
RUN"VOS
```

V-OS loads the kernel, switches off the ROMs and takes over the whole machine. It detects the memory expansion by itself.

- **First view:** the Desktop. You can change it in Setup, on the START VIEW line.
- **Language:** English. If there is a `SETUP.CFG` on the disc, it starts with the language and settings saved there.
- **Exit:** FILE > EXIT TO BASIC (CONTROL+Q in the views). It asks for confirmation and restarts the CPC.

The screen

V-OS uses the whole screen (overscan): 384×272 pixels in mode 1 (4 colors) or 192×272 in mode 0 (16 colors).

- **Menu bar** (top): the menus of the view or program, usually FILE or the program's menu, VIEW and HELP. On the right is the keyboard language (EN, EΛ, ...).
- **Bottom line:** in each view, the keys that work there.

Keys that work everywhere

Key	What it does
F1	Desktop
F2	Launcher
F3	Commander
F4	Setup
F5	opens and closes the desk accessory (ACCESSORY)
F6	the running programs, to choose one
ESC	opens the first menu. In the menus, ←/→ change menu, ↑/↓ choose an item, RETURN runs it, ESC closes
CONTROL+letter	the menu shortcuts (e.g. ^Q, ^L)

CONTROL+L	switch language (VIEW > SWITCH LANGUAGE)
CONTROL+K	the dialog with the keys (HELP > KEYS)
CONTROL+SHIFT	switches the keyboard between English and the language of the pack
TAB, SHIFT+TAB	move the focus between buttons. RETURN, ENTER or SPACE press the button that has the focus

F1–F6 are the f1–f6 keys on the CPC's numeric keypad.

The pointer and the mouse

- **With an AMX mouse:** the pointer moves with the mouse, and its button clicks. The mouse is switched on and set up in Setup (AMX MOUSE, AMX READING).
- **Without a mouse:** the pointer moves with the arrow keys, and COPY clicks. In views that use the arrow keys for choosing (Desktop, Launcher, Commander), the arrows choose files and programs instead of moving the pointer.

Most things work with both: one click on a file or program selects it, and a second click opens it.

Windows and dialogs

- **Dragging:** every window can be dragged by its title bar: the dialogs, the Desktop's FILES window, the desk accessory, and the windows of block programs. While you hold the button a dashed outline is shown, and the window is redrawn where you let go.
- **Dialogs:** wait for an answer. TAB and ←/→ change the button, RETURN presses it, and ESC chooses the last one (usually NO or CANCEL).
- **Text input** (e.g. renaming): you type at the end, DEL deletes, RETURN accepts and ESC cancels. You can also type in the language of the pack.

Discs and files

- **Format:** V-OS reads and writes AMSDOS discs (DATA or SYSTEM format) in A: and B:. BASIC can read its files too.
- **Names:** 8 characters plus 3 for the type (`NAME.TYP`).

Types you will come across:

Type	What it is
`.BIN`	a V-OS program (or a binary file)
`.VBN`	a V-OS program made with blocks
`.VBS`	a Studio program (its source)
`.OVL`, `.MSG`, `.LNG`	parts of V-OS itself: views, texts, languages
`.SYS`	system files (e.g. `VBRUN.SYS`, the Studio player)
`SETUP.CFG`	the settings

The settings, languages and views are read from A:, so the V-OS disc must stay there.

Error messages

Disc errors are shown in a dialog: DRIVE NOT READY, WRITE PROTECTED, FILE NOT FOUND, DISC FULL, DIRECTORY FULL, NAME ALREADY USED, INVALID NAME and others.

Two more concern programs:

- **NOT A V-OS PROGRAM:** the file does not have the header of a program.
- **CLOSE ANOTHER PROGRAM FIRST:** there is no room. See chapter 3.

2. The views

V-OS has four "views": ways to see the files and programs, and the settings. You switch view with F1–F4 or from the VIEW menu. Each view remembers which file was selected, and the next one selects the same file.

The common menus

The last two menus are the same in all views and in programs:

VIEW		HELP	
DESKTOP	F1	KEYS	^K
LAUNCHER	F2	ABOUT V-OS	
COMMANDER	F3		
SETUP	F4		
PROGRAMS (running)	F6		
SWITCH LANGUAGE	^L		

HELP > ABOUT V-OS shows the version (0.1 beta), the date of the build, the maker (Revive8bit, 2026) and the architect (Vasper).

Desktop, F1

Mode 1. A FILES window shows the files of a drive as icons, four in each row. The type is written inside the icon and the name below it, and a `.VBN` has a "play" mark next to it. On the right, two icons choose A: or B:. The title bar shows the disc format and the number of files, and the free space is shown at the bottom.

Key	What it does
arrows	choose a file (↑/↓ one row)
TAB	the other drive
RETURN, ENTER	open: a program runs, a file of a known type opens in its program, others show their info
^C	copy to the other drive
^D	delete (asks first)
^N	rename
^R	read the discs again
mouse	click on a file: select, again: open. Click on a drive: opens it

- **FILE menu:** OPEN, FILE INFO (^I), COPY TO OTHER DRIVE, DELETE, RENAME, READ DISCS AGAIN, EXIT TO BASIC.
- **The FILES window** can be dragged by its title and stays there, even when you switch view.
- **Opening by type** (also in the Commander): `.TXT` files open in NOTEPAD, `.MD` in MARKDOWN, `.PIC` and `.LYR` in PAINT, `.MUS` in MUSIC, which starts playing them. The program is looked for on the file's drive, then on A:, then on B:. If it is not found, "NO PROGRAM FOR THIS FILE" is shown. If the program has unsaved changes, it asks first.

Launcher, F2

Mode 0, 16 colors. Shows only the programs of one drive (`.BIN`, `.VBN`, `.BAS`), A: at first, as tiles in a 3×3 grid, with the next pages below. The page line starts with the drive (`A:`, `B:`). Every program that

has an icon shows it on its tile. The selected tile is white.

Key	What it does
arrows	choose (↑/↓ one row. Past the edge, the next or previous page)
RETURN, ENTER	runs the program. If it is already running, it resumes where it was
^A	loads it as the desk accessory (AS ACCESSORY). F5 then opens it
^R	read the disc again
TAB	the other drive (OTHER DRIVE). A program started from B: comes back to B:
mouse	click: select, again: run

FILE menu: RUN, AS ACCESSORY, OTHER DRIVE, READ DISCS AGAIN, EXIT TO BASIC.

Commander, F3

Mode 1, two panels with file lists: A: on the left, B: on the right. Each panel shows the format and the number of files in its title, and the free space at the bottom. The active panel has a dark title and a selection.

Key	What it does
↑/↓	choose (with SHIFT: one page)
TAB, ←/→	the other panel
RETURN, ENTER	open, as in the Desktop. The info (size, type, load address, the first bytes) with ^I
^C	copy to the other panel, of any size. If there is a file with the same name, it asks first
^D	delete
^N	rename
^R	read both discs again
mouse	click: select, again: open

FILE menu: OPEN, FILE INFO (^I), COPY TO OTHER PANEL, DELETE, RENAME, READ DISCS AGAIN, FORMAT DISC (DATA), EXIT TO BASIC.

> FORMAT erases all the files on the disc of the active panel. It asks first.

Setup, F4

Mode 1. One line for each setting. Changes take effect at once. When saved (RETURN or ^S) they are written to `A:SETUP.CFG` and also apply at the next start-up.

Setting	Values
START VIEW	DESKTOP, LAUNCHER, COMMANDER
LANGUAGE	ENGLISH, ΕΛΛΗΝΙΚΑ, FRANÇAIS, ESPAÑOL, DEUTSCH, ITALIANO
REPEAT DELAY	how long you hold a key before it starts repeating: SHORT, MEDIUM, LONG
REPEAT SPEED	SLOW, MEDIUM, FAST
AMX MOUSE	ON, OFF
AMX READING	PULSES or LEVELS: how the mouse is read. If the pointer does not move properly, try the other one
DESK COLOUR	BLUE, BLACK, GREEN, RED

Key	What it does
↑/↓	choose a setting
←/→	change the value (also a click on the line)
RETURN, ^S	save

Changing the LANGUAGE loads the language pack from A: at once. See chapter 4.

3. Programs

Starting and ending

A program starts in one of these ways:

- from the Launcher: select it and press RETURN, or click twice
- from the Desktop or the Commander: FILE > RUN
- from another program, such as Studio, which runs whatever you make

Each program has its own menu next to the common VIEW and HELP. It ends from its menu (usually BACK or STOP, CONTROL+Q) and returns to the view it was started from.

Several programs at once

V-OS programs are of two kinds:

- **Small, with a relocation table:** DEMO, MIXED and CLOCK, the web editor examples (PONG, CATCH, STARCAT, COUNTER), and whatever the SDK makes. They share 12K of memory, and as many as fit run together.
- **Programs that need all the memory:** STUDIO and whatever Studio makes (CALC, MINES, FLAP, TICKER and your own). Without a memory expansion they run on their own. If one is already running, you get "CLOSE ANOTHER PROGRAM FIRST".

When you go to a view (F1–F4) while a program is running:

- A small program stays in memory. You pick it up again from the Launcher (RETURN on it) or with F6, and it resumes exactly where it was.
- A program that has all the memory ends, unless there is a memory expansion.

F6 (VIEW > PROGRAMS) shows the list of running programs. You choose one with the arrows and RETURN. If none is running, you get "NO PROGRAM IS RUNNING".

With a memory expansion (e.g. 512K, which V-OS finds by itself at start-up), every program that needs all the memory gets its own bank. So STUDIO, CALC, DEMO and PONG can all run together.

In the background

A program behind another one, or behind a view, normally **freezes** and resumes when it comes back to the front.

Some programs ask to **keep running** in the background too, e.g. TICKER, which counts time. While there they do not draw or play sound. When they come back to the front, they have also counted the time they were away.

In Studio you choose this for your program with STUDIO > IN BACKGROUND.

The desk accessory (ACCESSORY)

A small program, such as CLOCK, can open in a window **on top of anything**: a view or a program.

1. In the Launcher, select CLOCK and choose FILE > AS ACCESSORY (^A).
2. From then on, **F5** opens and closes it, wherever you are. ESC also closes it.

While it is open, it gets the keys and clicks. Its window can be dragged by its title. When it closes,

whatever was underneath is redrawn.

The programs on the disc

Program	What it is
STUDIO	the block editor: you make programs on the CPC (chapter 5)
DEMO	the first V-OS program: a square that moves by itself. The arrows steer it and SPACE changes its color
MIXED	the "mixed screen": a band with 16 colors (mode 0) between mode 1 bands. A click shows the coordinates
CLOCK	the time since V-OS started. It also runs as a desk accessory
CALC	a calculator in a window with buttons, made in Studio (there is also `CALC.VBS`)
MINES	an 8×6 minesweeper with 8 mines, made in Studio. A click opens a cell, and NEW starts a new game
BRICKS	brick breaking: a paddle with ←/→, 32 bricks, 3 lives, levels with an ever faster ball. At GAME OVER, SPACE for a new game
VROUVES	tower defence, from the CPC game LET'S GO FOR VROUVES: 7 stages, from the Kafenio to the cemetery, with 10 waves each. It is on its own disc, `vrouves.dsk`: put it in B: and press TAB in the Launcher. Arrows move the cursor, 1–6 picks a defender, SPACE builds, S sells, N calls the next wave, P pauses. Points for every invader and for the villagers left at the end of a stage. The disc keeps `VROUVES.SAV`: CONTINUE starts from the stage you reached, HIGH SCORES shows the best five with their names. Written with the SDK (`myapps/vrouves`)
NOTEPAD	a text editor in mode 1, 48×22, for files up to 4K. NEW, OPEN (only `.TXT` files), SAVE, SAVE AS, OTHER DRIVE. Arrows, DEL, CLR and RETURN; a click places the cursor. The bottom line shows the line and the column. EDIT menu: COPY ^C, CUT ^X, PASTE ^V, SELECT ALL ^A; SHIFT and the arrows select. The clipboard (up to 1K) is shared by all programs (`myapps/notepad`)
PAINT	drawing in mode 0 with 16 colors: a 128×128 picture (`.PIC`), or 64×64 at 2× zoom in 3 layers (`.LYR`, NEW LAYERS). Tools: P pencil, B brush, L line, X box, F fill; TAB changes the color. Arrows and COPY draw without a mouse. Layers: 1–3 choose one, H hides or shows it, U and D move it up and down, E is the eraser (`myapps/paint`)
MUSIC	plays Arkos Tracker 3 songs (`.MUS`, made by `make songs`), with three volume bars. It is on its own disc, `music.dsk`, in B:. OPEN, SPACE plays and stops. The music goes on behind other views and programs (`myapps/music`)
MARKDOWN	an editor for Markdown files (`.MD`, up to 6K), like NOTEPAD. ^P shows the page: `#` headings, **bold**, *italic*, `code`, lists, quotes, lines, code blocks and underlined links, without the marks. The arrows scroll the page, RETURN or ^P go back to the text. It is on its own disc, `markdown.dsk`, with a `README.MD`. EDIT menu: COPY ^C, CUT ^X, PASTE ^V, SELECT ALL ^A; SHIFT and the arrows select. The clipboard (up to 1K) is shared by all programs (`myapps/markdown`)
SYSINFO	shows the V-OS and API versions and the free space on the discs in A: and B: (^R again). An example of `VOS_EXT` (`myapps/sysinfo`)
PONG	a ball and a paddle with the arrows, score and lives
CATCH	catch the falling stars
STARCAT	a cat with costumes that walks, a star, a melody from a list
COUNTER	a counter with +1, −1 and 0 buttons, in mode 1
FLAP	a bird that flaps its wings (costumes), and a star that changes costume every second
TICKER	counts seconds and keeps going in the background

- Block programs are `.VBN` files. They have a PROGRAM > STOP menu (CONTROL+Q).
- Those that also have a `.VBS` (CALC, MINES, FLAP, TICKER) open in Studio so you can look at

them or change them.

- `VBRUN.SYS` is not a program: it is the "player" that Studio needs.

4. Languages

V-OS speaks six languages. English is built in. The other five are **language packs**, files on the disk that bring three things: all the system texts, the keyboard, and the letters (font) of the language.

Language	File	Keyboard
English	built in (`ENGLISH.MSG`)	the CPC's English one
Ελληνικά	`GREEK.LNG`	Greek, with accents
Français	`FRENCH.LNG`	AZERTY, digits with SHIFT, dead keys ^ and ¨
Español	`SPANISH.LNG`	Ñ on the `:` key, dead keys ´ ¨ ` ^, and ¡ ¿
Deutsch	`GERMAN.LNG`	QWERTZ, Ü Ö Ä ß, dead keys ´ and `
Italiano	`ITALIAN.LNG`	è é ò à ù ì on their own keys

Switching language

- **Permanently:** in Setup (F4), the LANGUAGE line cycles through all the languages. Each one loads from A: at once, and the whole screen is redrawn in it. Saving (^S) keeps it for the next startup too.
- **Quickly:** CONTROL+L (VIEW > SWITCH LANGUAGE) switches between English and the language of the pack that is loaded.
- **Keyboard only:** CONTROL+SHIFT switches the keyboard between the English one and the pack's. The menu bar shows on the right which one is active (EN, ΕΛ, FR, ...).

The symbols `@ # [ ] { } \ |` are not on the packs' keyboards, because the CPC has no AltGr. Type them with the English keyboard.

Program texts

Programs have their own texts in **two languages**: English and, optionally, one more (today, Greek). A program appears in its second language only when the pack is in that language. Otherwise it appears in English. So, with French, Studio appears in English.

Making a new language

The language sources are in `lang/<code>/`:

- `messages.txt`: the system texts. The base list is defined by `lang/en/messages.txt`, and anything missing from a pack appears in English.
- `keymap.txt`: the keys that differ from the English keyboard.
- `pack.txt`: the code, the name, the character set and the font.

`tools/mklng.py` turns them into a `.LNG`, and the Makefile puts it on the disk. The Latin packs share the font `lang/latin1/font.txt`.

5. Studio: programs with blocks

STUDIO is a Scratch-style block editor that runs on the CPC itself.

- You make sprites and scripts out of blocks.
- The program runs at once with CONTROL+R.
- It becomes a normal V-OS program (`.VBN`) that also starts from the Launcher.

The screen

Studio uses a "split" screen, with two zones:

- **Top, in mode 0 (16 colors):**
  - **left, the stage** at half size
  - **right, the SPRITES panel:** the sprites with their color and position, the size of the selected one (W, H), the background and the program's mode (e.g. `STAGE 2 M1`), and the file name
- **Bottom, in mode 1:**
  - **left, the palette** with the blocks of one category (`< EVENTS >`)
  - **right, the script**, one line per block
  - **at the end, the status line**, with the keys that work where you are

TAB goes from the script to the palette, then to the stage, and back to the script.

Menus

STUDIO		EDIT
NEW	^N	DELETE LINE
OPEN (`.VBS`)	^O	+ SPRITE (up to 6)
SAVE	^S	− SPRITE
SAVE AS		BACKGROUND (the stage's color)
RUN	^R	SCREEN MODE (0 or 1)
IN BACKGROUND		
BACK	^Q	

- **RUN (^R):** saves `NAME.VBS`, writes `NAME.VBN` and runs it. The program stops from its own menu (PROGRAM > STOP, ^Q), and then Studio starts again with the same program open.
- **IN BACKGROUND:** asks whether the program keeps running when it is behind another one or behind a view, or pauses, which is the default.
- **SCREEN MODE:**
  - **Mode 0:** 16 colors, 192 pixels wide.
  - **Mode 1:** 4 colors (pens 0–3), 384 pixels wide, and more text.

The palette

←/→ change the category, ↑/↓ choose a block, and RETURN adds it below the script line. A click on a

palette block does the same.

A block that opens (FOREVER, REPEAT, IF) goes in together with its END, and is deleted together with it. A block with text asks for the text at once.

The script

Key	What it does
↑/↓	line
←/→	slot of the line (the slots are the values inside the block)
RETURN	with no slot selected: adds the palette block. On a text slot: change the text. On a key slot: waits for you to press the key
CLR	deletes the line
DEL	deletes the last digit of a number
mouse	click on a slot: selects it. Second click: changes its kind

The program is a list of lines:

- Each `WHEN …` starts a script, for a sprite or for the stage. The script runs up to the next `WHEN`.
- The indentation comes by itself from the blocks that open and close.
- Lines before the first `WHEN` are ignored.
- Anything left open is closed at the end of the script.

What goes in a slot

The status line always says what the slot accepts.

Value slot:

Key	Value
0–9, −	number, an integer from −32768 to 32767
A–H	variable. The same letter again: A1, A2 … A15, and back to A
X, Y	the position of the script's sprite
R and digits	RANDOM: a number from 1 to n
K and a key	KEY: 1 while it is held down, otherwise 0
T and a digit	TOUCHING: 1 when the sprite touches sprite n
Z	AT EDGE: 1 when the sprite is at the edge of the stage
P	BUTTON: the number of the last button pressed
U	COSTUME: the number of the sprite's costume (0: no costume)

Other kinds of slot:

- **Variable:** A–H, and the same letter again for A1…A15.
- **Comparison:** `<`, `=`, `>`. SPACE gives "none", and then IF only checks that the first value is not 0.
- **Sprite** (in the `WHEN` blocks): 0 the stage, 1–6 a sprite.
- **Key** (in `WHEN KEY`): RETURN and then the key.
- **Text:** RETURN, and you type up to 14 characters.
- **Variable group** (SAVE/LOAD VARS): A–H moves to the next group of eight (A-H, A1-H1 … A15-H15), and SPACE gives ALL.

The letters are read as keys, so they also work with the Greek keyboard.

The stage and the sprites

Key	What it does
arrows	move the selected sprite
1–6	select a sprite
C	next color
W, H	width, height
K	which costume it wears (none, 1, 2, ...)
E	draw its costume (a new one, if it has none)
click on the stage	puts the sprite there

A sprite without a costume is a colored rectangle.

Costumes

E opens the costume editor in place of the stage. It has a 16×16 grid with the 16 pens below it.

Key	What it does
arrows	cell
SPACE, RETURN	paints the cell
0–9, A–F	pen
.	transparent
E, TAB	done
click	on a cell: paints. On the row of pens: selects a pen

A program has up to 8 costumes, shared by all sprites. The size of a sprite with a costume is 16×16. On the stage, x is rounded to a whole byte (2 pixels in mode 0, 4 in mode 1).

All the blocks

In the "Block" column, the slots appear as in the script.

EVENTS

Block	What it does
WHEN START *sprite*	the script starts together with the program
WHEN KEY *key sprite*	starts when the key is pressed
WHEN CLICKED *sprite*	starts with a click on the sprite (or on the stage)
WHEN I RECEIVE *n sprite*	starts when message n is sent
SEND *n*	sends message n (n can also be a variable)

> A message starts **from the beginning** every script that waits for it, even if it was already running. That is why a script should not send a message to itself.

CONTROL

Block	What it does
FOREVER ... END	repeats forever
REPEAT *n* ... END	n times
IF *a comparison b* ... [ELSE ...] END	runs the lines only if the comparison holds
WAIT *n*	waits n frames (50 = one second)

WAIT UNTIL *a comparison b*	waits until the comparison holds
STOP	ends this script

MOTION

Block	What it does
GO TO *x y*	the sprite goes to x, y
CHANGE X *n*, CHANGE Y *n*	moves by n
SPEED *vx vy*	sets its speed
MOVE	one step at its speed
BOUNCE	at the edge of the stage, reverses the speed

LOOKS/SOUND

Block	What it does
COLOUR *n*	the rectangle's pen
COSTUME *n*	wears costume n (1–8. 0 changes nothing)
NEXT COSTUME	the next one, and after the last one the first again
SHOW, HIDE	
PRINT *value* AT *x y*	prints a number
NOTE *n* FOR *frames*	plays a note (MIDI number: 60 = C, 69 = A 440 Hz) and waits for it to finish

VARIABLES

Block	What it does
SET *variable* TO *value*	
CHANGE *variable* BY *value*	addition
DECREASE *variable* BY *value*	subtraction
MULTIPLY *variable* BY *value*	
DIVIDE *variable* BY *value*	integer division
REMAINDER *variable* BY *value*	the remainder of the division
SAVE VARS *group* TO *"name"*	writes the variables to a text file (see below)
LOAD VARS *group* FROM *"name"*	reads them back

WINDOWS

Block	What it does
WINDOW *"title" width height*	a window in the middle of the stage, like the V-OS dialogs. A new window takes the place of the previous one and its buttons
BUTTON *"text"* AT *x y*	a system button. It is pressed with a click or with TAB and RETURN. Buttons are numbered from 0, in the order they appear
WHEN BUTTON *sprite*	starts when any button is pressed. The BUTTON value (P) says which
TEXT *"text"* AT *x y*	writes text
LABEL *button* TO *"text"*	changes the text of a button

With a window open, TEXT, BUTTON and PRINT count from the top-left corner of its inside. The window is dragged by its title, together with its buttons.

Variables

There are 128 variables, shared by all scripts, named A–H, A1–H1, ..., A15–H15. At the start they are all 0, and they hold integers from −32768 to 32767.

Saving and restoring (SAVE VARS, LOAD VARS)

- `SAVE VARS ALL TO "GAME"` writes all 128 to the file `GAME.TXT`. The `.TXT` is added when the name has no type.
- With a group (`A-H`, `A1-H1`, ...) it writes only those 8.
- The file is text, one line per variable:

```
A=12 B=0 A1=-5
```

- `LOAD VARS ALL FROM "GAME"` changes the variables the file mentions. With a group, it changes only those that are also in the group. Lines it does not understand are ignored, and if the file is missing nothing changes.
- So a file can restore all the variables or just one group of eight, e.g. a game's saved positions.

Files

- **`NAME.VBS`:** the program as you edit it. OPEN opens it, and up to 5K is read.
- **`NAME.VBN`:** the program ready to run. RUN writes it, and it also starts from the Launcher. It is the player (`VBRUN.SYS`) together with the program. The player compiles the blocks to Z80 code when it starts, so programs run fast.

A `.VBS` can also be made from text on the PC, with `tools/mkvbs.py`. The syntax is described at the top of the file, and examples are `blocks/examples/*.vbt` (CALC, MINES, FLAP, TICKER, VARS).

Limits

Lines (blocks)	240
Sprites	6
Scripts (`WHEN` ...)	16
Variables	128
REPEAT in one program	64 (beyond that they share a counter)
Open blocks nested inside each other	16
Texts	32, up to 14 characters each
Costumes	8, 16×16
Buttons	64
Code	at least 4,000 bytes, even for the largest program. Whatever does not fit is ignored

Tips

- **Stage scripts:** the motion and looks blocks do nothing, and X, Y, TOUCHING and AT EDGE give 0.
- **TOUCHING:** holds only when both sprites are visible.

- **Timing:** scripts run "together". Each script gives up its turn at the end of every loop and at every WAIT. To make one script wait for another, use a variable and `WAIT UNTIL` (see MINES).
- **Ready-made examples:** CALC, MINES, FLAP and TICKER open in Studio (OPEN).

6. Blocks on the PC: the web editor and `vbc.py`

Besides Studio, which runs on the CPC, there is also a block editor for the PC. The language is the same, with a few extra blocks: lists, noise, sprite size, colored text.

The editor

Open `blocks/editor.html` in a browser. It is based on Blockly and comes in Greek or English.

- **Categories:** events, control, motion, looks, text, operators, sensing, variables, lists and windows.
- **Sprites:** rectangles or costumes. They are defined in a table, and the stage shows them in the CPC's colors. They can be dragged with the mouse.
- **Files:** it opens and saves `.vbp` files. They are JSON with the whole program, so they can also be written by hand.

Compiling

```
python3 blocks/vbc.py pong.vbp --dsk build/vos.dsk
```

`vbc.py` writes `build/NAME.ASM`, with a comment for each block. It assembles it with RASM into `NAME.VBN`, with a relocation table so that it runs alongside other programs, and with `--dsk` it puts it on the disk.

Option	What it does
`-o build`	output folder
`--asm-only`	only the `.ASM`
`--dsk build/vos.dsk`	also onto the disk

The `.vbp` format

The full description is at the top of `blocks/vbc.py`: events, commands, values, costumes, icon. In short:

```
{"name": "PONG", "mode": 0, "background": 0,
 "sprites": [{"name": "ball", "x": 90, "y": 100, "w": 6, "h": 6,
              "colour": 8, "vx": 2, "vy": 2, "visible": true}],
 "variables": {"score": 0},
 "scripts": [{"sprite": "ball", "when": {"event": "start"},
              "do": [{"block": "forever", "do": [{"block": "move"}, {"block": "bounce"}
```

- **Variables and lists:** they have names, not letters. Lists have a fixed maximum length.
- **Icon:** `"icon"` gives an icon for the Launcher.
- **A program with many similar parts:** a Python script can write it. See `tools/mkbricks.py`, which writes `bricks.vbp` with 32 bricks.

Examples

In `blocks/examples/`:

File	Program
`pong.vbp`	PONG

`catch.vbp`	CATCH
`starcat.vbp`	STARCAT: costumes, a tune from a list
`counter.vbp`	COUNTER: a window and buttons
`bricks.vbp`	BRICKS (written by `tools/mkbricks.py`)

The `*.vbt` files are Studio programs as text, for `tools/mkvbs.py`. See chapter 5.

Studio or web editor?

	Studio	**web editor**
Where	on the CPC	on the PC
File	`.VBS`	`.vbp`
Variables	128, A–H15	with names
Lists, noise, sprite size	no	yes
SAVE/LOAD VARS	yes	no
The program	needs all the memory	small, runs alongside others

7. The V-OS SDK: programs in assembly

With this SDK you write your own programs in Z80 assembly. They run inside V-OS like DEMO or CLOCK: they appear in the Launcher, the Desktop and the Commander, and they have their own menu, buttons, dialogs, sound and files. They also run alongside other programs.

Quick start

```
make new-app NAME=hello
```

This creates the folder `myapps/hello/` from `sdk/template/`. It has three files:

- `hello.asm`: the program. The arrows move a box, there is a BEEP button, a seconds counter, and its own menu (ABOUT, BACK) next to VIEW and HELP.
- `texts.txt`: its texts, in English and Greek.
- `icon.txt`: its icon, 16×16, for the Launcher.

```
make app NAME=hello
```

This builds `build/HELLO.BIN` and puts it on the disk `build/vos.dsk`.

```
make run
```

This opens the disk in Caprice32. HELLO is in the Launcher, on the last page.

- Plain `make` builds all the programs in `myapps/` and puts them on the disk, together with V-OS.
- The name has 1–8 lowercase letters or digits and starts with a letter. It cannot be the name of a V-OS program (e.g. demo).
- You need RASM, Python 3 and iDSK, as for all of V-OS.
- In Caprice32 you run the program yourself, from the Launcher.

A bigger program:

- Put more `.asm` files in its folder (e.g. `art.asm` with sprites) and bring them in with `include "art.asm"`. The build finds them, and rebuilds the program when they change.
- Whatever you put in the subfolder `disc/` (e.g. `disc/LEVEL1.DAT`) goes onto the disk as it is, for `VOS_LOAD`.
- With a subfolder `owndisc/`, the program also gets a disk of its own, `build/<name>.dsk`, with the `.BIN` and the files of `owndisc/`. It stays off `vos.dsk`. You put it in B: and run it from the Launcher (TAB, the other drive). Which drive it was started from, so that it loads its files from there, is in `VOS_SELF`: `db drive`, then its name (11 bytes).
- **A file to open (version 14):** when RETURN in the Desktop or the Commander opens a file in your program, `VOS_OPEN` (`#B000`) holds `db 1, drive, 11-byte name` (0: none). Read it in `init`, with A = 0 and also with A = 2 (the program was already running), then write 0 there. The types and their programs are in the kernel's `ft_open` table (`src/kernel/files_ext.asm`).
- A program with a relocation table (like those of the SDK) can be up to 12K. For the rest, load files into the data bank.

How a program works

A program is a `.BIN`, built for `VOS_APP_ADDR` (`#4000`). It starts like this:

```
        include "api/vos_api.inc"
        org VOS_APP_ADDR
        jp init                 ; +0
        jp event                ; +3  HL -> the event
        jp redraw               ; +6
        db "VA", VOS_API_VERSION
        include "app_icon.asm"  ; +12 the icon (optional)
```

- **`init`** sets mode, palette, menu and buttons, turns on `VE_FRAME` if needed, and draws everything.
  - A = 0: a fresh start.
  - A = 2: the program returns from a view or another program. Its state is as it left it; it sets the above again and draws.
- **`event`** is called for each event and returns immediately. A program never waits on its own (no wait loop). For motion or time, `VOS_TICKS` with A = 1, and then a `VE_FRAME` arrives 50 times a second.
- **`redraw`** redraws everything, e.g. when the language changes or the desk accessory closes.
- **Exit:** `VOS_EXIT`, which returns to the view that started it.

The order inside `event`

```
event:  ld a,(hl)
        cp VE_FRAME : jp z,frame        ; the frames
        push hl
        call VOS_UI_EVENT               ; menus, buttons, CONTROL+letter
        pop hl
        or a : jr nz,command            ; A = the command
        call VOS_MENU_OPEN : ret nz     ; menu open: the keys are its own
        ...                             ; your own keys and clicks
command: ...
        jp VOS_COMMAND                  ; whatever is not yours (VIEW, HELP)
```

While a menu is open, do not draw. Otherwise you will draw over it. On `VE_FRAME`, update the state, but draw only if `VOS_MENU_OPEN` gives Z. Dialogs (`VOS_DIALOG`, `VOS_INPUT`, `VOS_FILE_PICK`) have their own loop, so no event arrives while they are open.

The events

`HL` → 8 bytes: `db type, code, char, mods; dw x, y`.

type	what	code / char / x, y
`VE_KEY_DOWN` 1, `VE_KEY_REPEAT` 3, `VE_KEY_UP` 2	key	code = key number (`VK_*`, keyboard row·8 + bit), char = the character in the pack's language
`VE_PTR_DOWN` 5, `VE_PTR_UP` 6, `VE_PTR_MOVE` 4	pointer	x, y in pixels of the mode
`VE_FRAME` 8	frame (with `VOS_TICKS`)	—
`VE_LANG` 7	the keyboard language changed	—

Three kinds of program

- **With a relocation table (the SDK default):** `make app` builds it twice, at `#4000` and at `#4100`. `tools/mkreloc.py` adds the table from the difference.
    - It runs alongside other programs, in the 12K from `#5000`, and continues where it stopped (F6 shows the running programs).
    - Its data must be inside its own bytes (labels, `ds`), never at fixed addresses.
    - If `mkreloc.py` gives an error, an address is used somewhere in a way that does not survive relocation. E.g. `ld a,label & 255`, or only the low byte of an address.
- **Without a table:** it takes the whole bank (up to 16K, code and data), but runs only on its own, unless there is a memory expansion. This needs your own rule in the Makefile (see `STUDIO.BIN`).
- **Desk accessory:** up to 4K, with a table. It opens with F5 on top of anything, and `init` gets A = 1. See `apps/clock/clock.asm` and the description in `src/api/vos_api.inc`.

Rules

- **Registers:** calls corrupt all registers except IX. Arguments go in A, BC, DE, HL (IX for `VOS_RECT` and `VOS_BLIT`). Results return in A or HL. On failure, carry is set and A = error number, which `VOS_ERROR` displays.
- **Memory:** do not page another bank in at `#4000`, because the program will vanish from under you. For 16K more there is the data bank (`VOS_BANK_GET`, `VOS_BANK_PUT`, `VOS_DATA`). It is shared with views and other programs, and keeps its data only while the program is running.
- **Coordinates:** x in pixels of the mode (mode 0: 0–191, mode 1: 0–383), y from 0 to 258. The menu bar takes y 0–11.
- **RASM:** labels are not case-sensitive (`BOX` and `box` are the same). Hex is written with `#` and not with `&`.

The data formats

- **Menu bar** (`VOS_SET_BAR`): `db count; dw menu, ...`. The shared VIEW and HELP come from `VOS_MENU_VIEW` and `VOS_MENU_HELP`.
- **Menu:** `db title, count` and for each item `db text, command, letter, 0`.
    - text: `VOS_SEPARATOR` for a line
    - command: 1.., what `VOS_UI_EVENT` returns
    - letter: CONTROL+letter shortcut, or 0
- **Buttons** (`VOS_SET_WIDGETS`, then `VOS_DRAW_WIDGETS`): `db count`, and for each button `dw x, y; db width in characters, text, command, 0`.
- **Dialog** (`VOS_DIALOG` → A = button, 0 the first, ESC gives the last): `db title, text, number of buttons (1-3), button, ...`. For buttons, `VM_OK`, `VM_YES` and `VM_NO`.
- **Texts** (`VOS_TEXTS`): made by `tools/mktexts.py` from `texts.txt`, with lines `NAME | English | Greek`.
    - It gives `T_NAME` (`VT_FIRST` = `#C0` and up) and the table `texts:`. Up to 64 texts.
    - A `|` with no spaces around it starts a new line in dialogs.
    - Texts go wherever a "text" is asked for (menus, buttons, dialogs). With `VOS_MSG` you get their address for `VOS_PRINT`.

- **Palette** (`VOS_PALETTE`): 17 firmware colors (0–26): 16 pens and the border. In mode 1, pens 0–3 count.
- **Rectangle** (`VOS_RECT`, IX →): `dw x, y, w, h; db pen`.
- **Icon:** 16 lines of 16 characters in `icon.txt`. Each character is a pen in hex (mode 0, Launcher palette) or `.` for transparent.
- **File names** (`VOS_LOAD`, `VOS_SAVE`, `VOS_DELETE`, `VOS_RUN`): 11 bytes, `"NAME EXT"`. Drive 0 = A:, 1 = B:.

The API (version 16)

Call	Input → output	What it does
`VOS_EXIT`	—	end, back to the view that started it
`VOS_TICKS`	A = 1 / 0 (+ `VOS_PTR_KEYS`)	`VE_FRAME` 50 times a second, or none. With `VOS_PTR_KEYS` (2) the arrows and COPY also move the pointer and click
`VOS_FRAMES`	→ HL (and A bits 16–23)	frames since startup
`VOS_RANDOM`	→ A	0–255
`VOS_COMMAND`	A = command	the shared commands of VIEW and HELP (6–19). Number your own commands from `VC_APP` (32) up, or a VIEW or HELP item runs one of yours
`VOS_ERROR`	A = error	error dialog
`VOS_MODE`	A = 0 / 1	mode 0 (192×259, 16 pens) or 1 (384×259, 4 pens)
`VOS_PALETTE`	HL → 17 colors	
`VOS_CLS`	A = pen	clears the screen
`VOS_PLOT`	HL = x, DE = y, A = pen	one pixel
`VOS_RECT`	IX → rectangle	filled rectangle
`VOS_COLOURS`	A = text pen (+ `VOS_BOLD`, `VOS_ITALIC`, `VOS_UNDERLINE`), C = background	for `VOS_PRINT` and `VOS_CHAR`. The styles are made from the same font, for every language
`VOS_PRINT`	HL = x, DE = y, BC → text, 0	
`VOS_CHAR`	HL = x, DE = y, A = character	
`VOS_MSG`	A = text → HL	the address of a text (`T_*` or a system one)
`VOS_SET_BAR`, `VOS_DRAW_BAR`	HL → menu bar	
`VOS_UI_EVENT`	(the current event) → A	menu, button or shortcut command, 0 for none
`VOS_DIALOG`	HL → dialog → A	
`VOS_INPUT`	A = title, HL → text, C = maximum → carry if cancelled	text input (needs C + 2 bytes)
`VOS_SET_WIDGETS`, `VOS_DRAW_WIDGETS`	HL → buttons	
`VOS_MENU_VIEW`, `VOS_MENU_HELP`	→ HL	the shared menus

`VOS_CATALOG`	A = drive → A = files, HL → list	16 bytes each: 11 the name, at +12 the size in K. With A = drive + `VOS_CAT_EXT` and HL → `"EXT"`, only the files with that extension
`VOS_FILE_PICK`	A = title → A = file, carry if cancelled	picker dialog, after `VOS_CATALOG`
`VOS_LOAD`	A = drive, HL → name, DE = where → BC = length	
`VOS_SAVE`	A = drive, HL → name, DE = from, BC = length	binary file, replacing the old one
`VOS_DELETE`	A = drive, HL → name	
`VOS_TEXTS`	HL → text table	the program's texts
`VOS_MENU_OPEN`	→ NZ if a menu is open	
`VOS_KEY_HELD`	A = key → NZ while pressed	for games
`VOS_BLIT`	IX → `dw x, y; db bytes, lines; dw source; db source stride`	image: pairs (mask, pixels)
`VOS_SOUND`	A = channel 0–2, HL = period (bit 15: noise), C = volume 0–15, B = frames	a note; 440 Hz is period 142
`VOS_SPLIT`	HL → `dw y1, y2; db 8 colors` (0 = turns off)	mixed screen: mode 0 between y1 and y2
`VOS_RUN`	A = drive, HL → name	runs another program, and then this one starts again
`VOS_KEEP`	→ HL → 16 bytes	memory that stays between programs
`VOS_BANK_GET`, `VOS_BANK_PUT`	HL = from, DE = to, BC = bytes	copy from and to the data bank
`VOS_BACKGROUND`	A = 1 / 0	keeps running or freezes when behind another
`VOS_DRAG`	HL = x, DE = y, BC = w, A = h → carry: new HL, DE	dragging a window by its title
`VOS_FIT`	same → HL, DE	brings a window inside the screen
`VOS_EXT`	A = a `VX_*` call, BC, DE, HL as it wants → A, HL, C, DE, carry	the calls after the table (version 15): `VX_VERSION` → A = API, HL = V-OS version (BCD). `VX_FREE` C = drive → A = free K. Unknown: carry `VX_CLIP_PUT` HL = text, BC = bytes → the clipboard (up to 1K). `VX_CLIP_GET` DE = where, BC = room → HL = size (version 16).

Full details, e.g. on the background, the accessory and `VOS_SPLIT`, are in `src/api/vos_api.inc`. For examples, see `apps/demo/demo.asm` (mode 0, motion), `apps/clock/clock.asm` (accessory) and `apps/mixed/mixed.asm` (mixed screen). A whole game with sprites, a stage from a file and sound is `myapps/vrouves/`.

An AI assistant writing a program should read `AI.md`: the same rules, shorter, with the traps of RASM and of the data bank.

Testing

`make test-sdk` builds the template as TEMPLATE.BIN and opens it from the Launcher in Caprice32, with key presses defined by `src/kernel/test_script.asm` (script 15). It takes screenshots in `build/shots`: its screen, the box after it moves, its menu, the ABOUT dialog, and the return to the Launcher.

For your own automated test, copy script 15 and the `test-sdk` target of the Makefile.

8. Developing V-OS

For those who work on V-OS itself. Programs *for* V-OS are described in the SDK.

What you need

- RASM, the assembler
- iDSK, for the disk images
- Python 3, for the tools. `make docs` also needs `markdown` and `pymupdf`.
- GNU make 4.3 or newer
- for testing, Caprice32 (snap `caprice32`)

Build

Command	What it does
`make`	`build/vos.dsk`: the kernel, the views, the languages, all programs and `myapps/*`
`make run`	opens the disk in Caprice32 (`RUN"VOS"`)
`make shot`	starts without a window and takes a screenshot
`make docs`	the manual as PDFs: `docs/V-OS.pdf` (Greek) and `docs/V-OS-en.pdf`, `-fr`, `-es`, `-de`, `-it`
`make windows`	`build/V-OS-windows.zip`: `V-OS.exe` for Windows, with CPCEMU 3.0.2 (CPC 6128, 576K, AMX mouse, M4 board), `vos.dsk` in A: and `vrouves.dsk` in B:. V-OS starts by itself. Needs WSL and the portable CPCEMU zip (`CPCEMU_ZIP=…`)
`make linux`	`build/V-OS-linux.tar.gz`: the `V-OS` script for Linux (x86_64), with CPCEMU 2.5 (3.0.2 is not out for Linux) and the same settings and discs. Needs SDL2 on the system and CPCEMU's Linux tar.gz (`CPCEMU_TGZ=…`)
`make songs`	the songs of Arkos Tracker 3 as `.MUS` files in `myapps/music/owndisc` (`tools/mkmusic.py`; needs Arkos Tracker 3, `ARKOS=…`)
`make new-app NAME=x`, `make app NAME=x`	your own program (SDK)
`make clean`	deletes `build/`

Repo layout

Folder	Contents
`src/vos.asm`	the root of the build: loader and kernel, writing to the disk
`src/boot/`	the loader that AMSDOS runs at `#4000`, RAM expansion detection
`src/kernel/`	vectors, interrupts, file system, views, API (`api.asm`), settings, kernel extension (`ext.asm`), test scripts (`test_script.asm`)
`src/drivers/`	Gate Array, CRTC, keyboard, AMX mouse, FDC, AY
`src/gfx/`	overscan graphics, text, mixed screen
`src/ui/`	menus, buttons, dialogs, text input, file picker
`src/shell/`	the views as overlays: Desktop, Launcher, Commander, Setup
`src/api/` `vos_api.inc`	the API for programs
`lang/`	the languages (texts, keyboards, fonts)

`apps/`	the programs in assembly: DEMO, MIXED, STUDIO, CLOCK, and their icons
`blocks/`	the web editor, `vbc.py`, `runtime.asm`, the Studio player (`player.asm`), examples
`sdk/`	the SDK: guide and program template
`tools/`	tools: fonts, languages, texts, icons, relocation, `.VBS` from text, screenshots, PDF
`docs/`	this manual in Greek (`manual/`) and in English, French, Spanish, German and Italian (`manual/en/`, `fr/`, `es/`, `de/`, `it/`), the PDFs, the requirements (`REQUIREMENTS.md`)

Memory, in brief

Area	What it holds
`#0000-#3DFF`	the kernel. In page zero, after the restart vectors, the API table at `#0108`
up to `#4000`	the kernel stack
`#4000-#7FFF`	the bank "window"
`#8000-#FFFF`	the 32K overscan screen. In its gaps, kernel buffers

The banks that go into the window:

Bank	What it holds
4	file and load buffer, and the data bank of the running program
5	the view at `#4000-#4FFF`, and from `#5000` the programs with a relocation table (or one program without a table in the whole bank)
6	whatever the menus and dialogs hide, and from `#7000` the desk accessory
7	the language pack, the Commander lists, the English texts, and from `#7000` the kernel extension

With a memory expansion, its banks take programs. The details, address by address, are in `docs/REQUIREMENTS.md` (§4.2 and in each "Υλοποίηση Μ..." (implementation) section).

Conventions

- **RASM:** hex is written `#C000` and not `&C000`, because RASM reads `&` as AND. Labels are not case-sensitive.
- **Comments:** in English, short, as in the existing code. Each file starts with a block that says what it does.
- **Documentation:** in Greek. `MILESTONES.md` (English), `MILESTONES.el.md` (Greek) and `docs/REQUIREMENTS.md` have a section for each milestone. The manual is written in Greek and translated into the other five languages of V-OS.
- **Texts:** the system texts are in `lang/*/messages.txt`, and the programs' texts in `texts.txt` (for `tools/mktexts.py`).

Testing

The tests run V-OS in Caprice32 without a window and take screenshots in `build/shots/<delay>/`.

How they work:

- Caprice32 without a window cannot press arrows, COPY or f keys. That is why the key presses are made by a special build: a `-DTEST_SCRIPT=n` puts script n into `src/kernel/test_script.asm`, a table of keys per frame.
- Each `make test-…` target builds a disk with its script and takes a screenshot at each delay in `…_SHOTS`.
- The delay is measured in "Z" presses: about 2.5–3.5 frames each. The mapping is not fixed,

so you catch a moment with two or three delays around it.

- Once V-OS writes to the disk, Caprice32 does not close by itself, and the test ends with a timeout. This is expected in test-vars, test-costume and others.
- You look at the screenshots yourself, e.g. combined into one image with `ffmpeg … hstack/vstack`.

Target	What it tests
`test-input`	keys, Studio: blocks, files, running
`test-calc`	CALC with clicks, then `CALC.VBS` in Studio
`test-costume`	costume in Studio, running, reopening
`test-acc`	the clock as a desk accessory
`test-multi`, `test-switch`, `test-expand`, `test-behind`	several programs, F6, memory expansion, background
`test-drag`	dragging windows
`test-langs`, `test-lang`	the languages, saving the language
`test-mines`, `test-bricks`	the games
`test-vars`	the 128 variables, SAVE/LOAD VARS
`test-sdk`	the SDK template
`test-vrouves`	VROUVES from its disc in B:, through the Launcher (TAB): building, wave 1, the funds, HIGH SCORES
`test-notepad`	NOTEPAD: typing, DEL, the save dialog, OPEN with only `.TXT` files
`test-paint`	PAINT: line, box, fill, the menus, then the layers (hiding one)
`test-about`	HELP > ABOUT V-OS
`test-music`	MUSIC from `music.dsk` in B:: a song plays, and goes on behind the Launcher
`test-markdown`	MARKDOWN from `markdown.dsk` in B:: `README.MD` opened, the page with ^P, a screen down, back to the text
`test-open`	opening by type in the Commander: `README.MD` in MARKDOWN from B:, a `.TXT` in NOTEPAD from A:
`test-sysinfo`	SYSINFO from the Commander: version, API, free K on A: and B:, ^R, ^Q
`test-clip`	the clipboard: selecting with SHIFT, ^C and ^V in NOTEPAD, ^A, ^X, ^V, then ^V in MARKDOWN
`test-amsdos`, `test-reboot`, `test-system`	V-OS files in BASIC, reboot, SYSTEM disk

New script:

1. Add an `IF TEST_SCRIPT == n` at the start of the chain in `test_script.asm`, with `ELSE` before the next one and one more `ENDIF` at the end.
2. Add a target to the Makefile, like `test-sdk`.

Each key is written as `dw from, to : db row, bit` of the keyboard. The key number is row·8 + bit, e.g. RETURN = 18 = row 2, bit 2.

Without an emulator: a program's code also runs on a Z80 inside Python (the `z80` package). You load the `.BIN` at `#4000` and make the API calls return immediately (or do whatever is needed, e.g. files in a dictionary). Then you call init and event and read the memory. It is much faster than screenshots for errors in numbers, files and jumps. The label addresses come from `rasm … -s -sl` (a `.sym` file).

Every milestone

1. Code, and a test in Caprice32: new or existing, with screenshots.
2. The tests the change touches.
3. A section in `MILESTONES.md` and `MILESTONES.el.md`, and a "Υλοποίηση Μ..." (implementation) section in `docs/REQUIREMENTS.md`.
4. **Update the manual in all six languages** (`docs/manual/` in Greek, `docs/manual/<code>/` for en, fr, es, de, it, and `sdk/README.md`, `sdk/README.<code>.md` if the API changes), then `make docs` for all the PDFs.
5. Commit and push.